Statistical Model Building

Presented by Dr Kathrin Schemann
Senior Statistical Consultant
Sydney Informatics Hub
Core Research Facilities

sydney.edu.au/sydney-informatics-hub

THE UNIVERSITY OF SYDNEY

Slides available here

Acknowledging SIH

- All University of Sydney resources are available to researchers free of charge. The use of the SIH services including the Artemis HPC and associated support and training warrants acknowledgement in any publications, conference proceedings or posters describing work facilitated by these services.

- The continued acknowledgment of the use of SIH facilities ensures the sustainability of our services.

Suggested wording for use of workshops and workflows:

- *"The authors acknowledge the Statistical workshops and workflows provided by the Sydney Informatics Hub, a Core Research Facility of the University of Sydney."*

What is a workflow?

- Every statistical analysis is different, but all follow similar paths. It can be useful to know what these paths are.

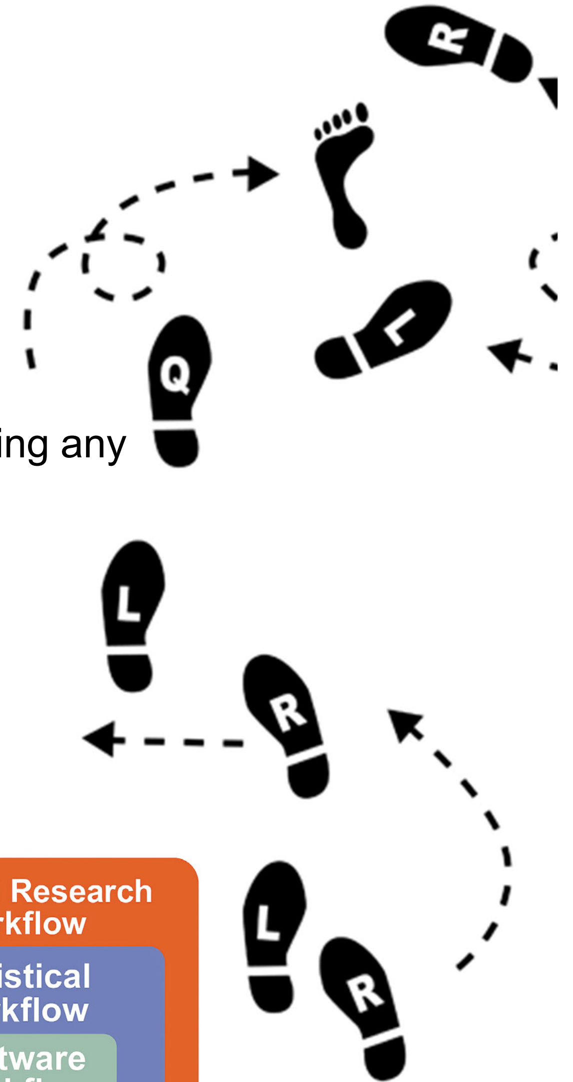
- We have developed practical, step-by-step instructions that we call ‘workflows’, that can you can follow and apply to your research.

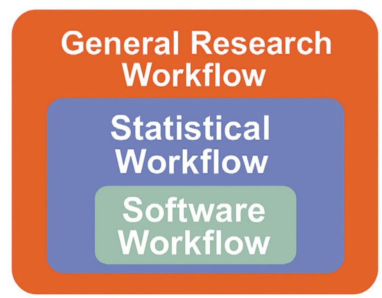
- We have a **general research workflow** that you can follow from hypothesis generation to publication.

- And **statistical workflows** that focus on each major step along the way (e.g. experimental design, power calculation, model building, analysis using linear models/survival/multivariate/survey methods).

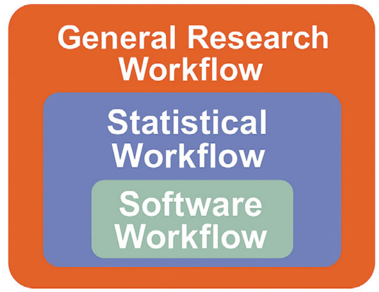
Statistical workflows

- Our **statistical workflows** can be found within our workshop slides.
- **Statistical workflows** are software agnostic, in that they can be applied using any statistical software.
- To access these statistical workflows and more, visit our Workshops and Workflows page.
- The broader aim of our workshop training is to increase understanding and application of statistical concepts, to facilitate higher impact research.





Software workflows



- There may also be accompanying **software workflows** that show you how to perform the statistical workflow using particular software packages (e.g. R or SPSS). We won't be going through these in detail during the workshop. If you are having trouble using them, we suggest you attend our monthly Hacky Hour where SIH staff can help you.

- Our software workflows contain:
  - R code and comments.
  - SPSS syntax as well as screenshots of the point and click procedures and written methods.
  - Screenshots of the point and click procedures and written methods for other bespoke software.

Using Artificial Intelligence (AI) responsibly

- USyd supports the responsible use of AI in research, aiming to balance innovation with academic integrity. USyd publishes its AI guardrails and endorsed AI tools, and currently allows Microsoft Copilot and Cogniti to be used with public and protected data. Highly protected data must never be entered into these tools.

- The Statistical Consulting Unit (SCU) also supports AI use, but like with any tool, it should be used responsibly. The SCU recommends:
  - Being careful with handling sensitive information.
  - Verifying the accuracy of all AI-generated content.
  - Ensuring the output generated is not false, misleading, or misrepresenting findings.

- This aligns with the Statistical Society of Australia's statement on the use of generative AI in statistics and data science.

Use AI to enhance, not replace statistical collaboration – it cannot match the nuance of a statistician who understands the research question, study design and constraints.

Using AI responsibly: Tips and tricks for statistical analysis

Ask detailed questions: Align each prompt to a specific step in a statistical workflow of your choice. E.g. for linear models, see below:

Be specific: State your outcome, predictors and how they are measured (numerically or categorically).

Protect your data: Avoid uploading data into AI tools. Instead, describe data structure, or upload simulated data or dummy tables.

Validate code: Check the code is statistically correct and using up to date packages and syntax. Understand what code does before running it.

Think critically: Always review, question and verify any content produced by AI.

Model Fitting Workflow

Step 0) Clean and check data.

Step 1) Pick a suitable model to fit to the data via Exploratory Data Analysis (EDA).

Step 2) Fit the Model.

Step 3) Check Model Assumptions via Diagnostics: Residual Analysis.

Step 4) Goodness of Fit: Plots and Statistics.

Step 5) Interpret Model Parameters and reach a conclusion.

Step 6) Reporting.

Understanding: Before relying on AI, make sure you understand the statistical foundations yourself. Or you won't know when it's wrong.

Check assumptions: AI rarely checks statistical assumptions unless explicitly asked. Check model diagnostic plots.

Statistical method should match study design: E.g. is your model accounting for clustering or repeated measures?

Why, not just what: Ask AI to explain why certain methods are appropriate for your data.

Using AI responsibly: Tips and tricks for R coding

Tip 1: Use **headings and subheadings,** as well as the **document outline** in R Studio to break down the problem into a workflow and think systematically.

- The example below uses subheadings to outline a workflow for data cleaning and processing.
- Breaking the problem into smaller parts leads to simpler questions for AI, that are easier to check.

```
# ... ----
# 3 DATA CHECKING AND CLEANING ----
## 3.1 Unique Identifier ----
# Expecting a unique identifier in the data export.
# We should not have duplicate values as there is one response per respondent.

unique.id <- combined_all # creating a save point!
names(unique.id) # check for the correct variable name and update the code below

# CHECK: the unique identifier is unique
table(duplicated(unique.id$UID))
```

- 0 Setup
- 1 Import Data
  - 1.1 Read Data
  - 1.2 Import checking
- 2 Combining data files
  - 2.1 Joining
  - 2.2 Bind rows
- 3 Data Checking and Cleaning
  - 3.1 Unique Identifier
  - 3.2 Data of Interest
  - 3.3 Unexpected characters
  - 3.4 Convert to numeric
  - 3.5 Create Factors
  - 3.6 Check Factor Levels
  - 3.7 Missing Data
  - 3.8 Duplicated Data
  - 3.9 Flatliners
  - 3.10 Outliers
  - 3.11 Final clean data
- 4 Create variables
  - 4.1 Top box

Tip 2: Under each heading/subheading write out the **purpose of the section**. Include AI prompts, as well as ideas for later (coding tends to be iterative).

Using AI responsibly: Tips and tricks for R coding

Tip 3: Every section has a **check**!

- This can be as simple as just having a look at the output, dimensions, or crosstabulations.
- Keep in mind you are responsible for your work and need to ensure your code does exactly what you want it to do. This is especially important when using AI as it can hallucinate!

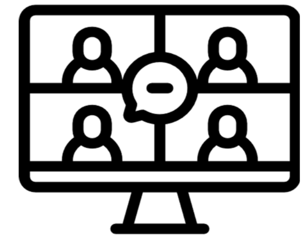
E.g. Should the number of rows change after joining a second data file (number of participants)?

```
> # CHECK: no additional rows
> nrow(raw_data_metro) # original n = 100
[1] 100
> nrow(combined_metro) # combined file should be the same!
[1] 100
```

E.g. Did all values get recoded correctly (were any NAs generated)?

```
> # CHECK: crosstab of original variable and new numeric variable
> # Looks good
> convert.numeric |>
+   count(Q3, Q3_numeric) |>
+   arrange(Q3_numeric)
                          Q3 Q3_numeric  n
1          Strongly Disagree          1 15
2                   Disagree          2 45
3 Neither Agree or Disagree          3 65
4                      Agree          4 42
5             Strongly Agree          5 27
```

How to use our workshops



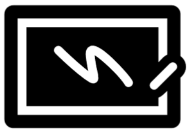
Workshops developed by the Statistical Consulting Team within the Sydney Informatics Hub form an integrated modular framework. Researchers are encouraged to choose modules to **create custom programs tailored to their specific needs**. This is achieved through:

- Short 90-minute workshops, acknowledging researchers rarely have time for long multi day workshops.
- Providing statistical workflows appliable in any software, that give practical step by step instructions which researchers return to when analysing and interpreting their data or designing their study e.g. workflows for designing studies for strong causal inference, model diagnostics, interpretation and presentation of results.
- Each one focusing on a specific statistical method while also integrating and referencing the others to give a holistic understanding of how data can be transformed into knowledge from a statistical perspective from hypothesis generation to publication.

For other workshops that fit into this integrated framework, refer to our training link page under statistics, found at [Workshops and training]

During the workshop

- Ask short questions or clarifications during the workshop (either by Zoom chat or verbally). There will be breaks during the workshop for longer questions.



- Slides with this blackboard icon are mainly for your reference, and the material will not be discussed during the workshop.

- Challenge questions will be encountered throughout the workshop.

Types of statistical models

Any “regression type” model, e.g.

- Linear Model (LM - numeric outcome variable)
- Generalised Linear Model (GLM – e.g. binary or count outcome variable)
- Linear Mixed Model (LMM – LM with a random effect, e.g. repeated measures)
- Generalised Linear Mixed Model (GLMM – GLM with random effect)
- Survival analysis (e.g. Cox semi-parametric regression; parametric regression)
- Structural Equation Modelling
- Bayesian Modelling
- Models of spatial data

Modelling is a very important part of quantitative research work.

A question for you:

What is your experience level with regression-type modelling?

a) No theoretical or practical experience.
b) Some theoretical experience only - from coursework, Linear Models WS)
c) Some practical experience - I have run a model for my research.
d) Experienced - I have run a number of models on different data.
e) Very experienced - I use models routinely.

What is a model?

Real world data

- A sample of reality
- Contains Information
- Contains Noise

Statistical model

- Separates information from noise
- Simplification of real world
- Low-dimensional summary of a dataset

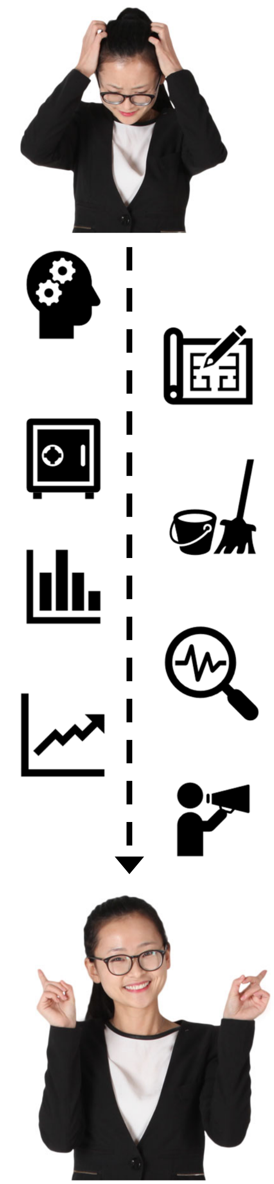
Inferences

- Describe relationships between variables
- Make predictions

"All models are wrong, but some are useful“ – George Box

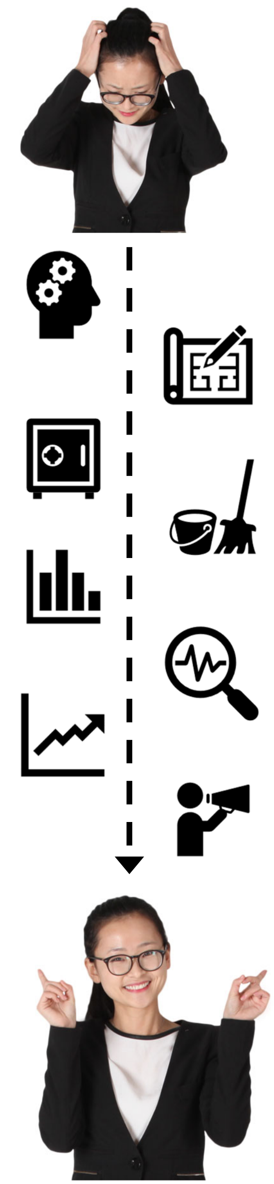
From *Research Essentials – Analysing your Data General Research Workflow*

1. Determine the Research Question
2. Study design and Analysis Plan (sampling, power, ethics approval)
3. Collect/Store Data
4. Data cleaning
5. Exploratory Data Analysis (EDA)
6. Data Analysis
7. Predictive modelling
8. Publication



From *Research Essentials – Analysing your Data*
General Research Workflow

1. Determine the Research Question
2. Study Design and Analysis Plan (sampling, power, ethics approval)
3. Collect/Store Data
4. Data cleaning
5. Exploratory Data Analysis (EDA)
6. Data Analysis (interpretative)
7. Predictive modelling
8. Publication



Fit a single model- LM1/2/3, Survival analysis

1. Check model assumptions
2. Check goodness-of-fit
3. Interpreting Model Parameters and reach a conclusion

Step 6: Data Analysis for Interpretation

Iteratively fit models - Model Building

- Pick predictors to fit and a suitable modelling method based on EDA
- Iteratively fit and assess models using predetermined strategy until no more criteria are met.

1. Determine the Research Question - what is the aim of the model?

Interpretable models seek to gain knowledge – we want to interpret the relationships between predictors + outcome:

- Need to get the most precise estimates
- Need to consider the level of subject matter knowledge/theory:

  - **Causal:** Fitting an a priori defined theoretical model – needs strong theory e.g. controlled experiment – aim is hypothesis testing/ causal inference
  - **Descriptive:** Fitting different models to create hypotheses about associations (weak theory)
  - **Predictive:** Getting most accurate predictions of future observations
    - Interpretable – test hypotheses about predictors that may be modifiable; or
    - Non-interpretable - if interested in prediction only, may use some Machine Learning approaches without making assumptions/ interpretations

Slide 19

KS1 Correct for model? Or just EDA? Check article

Kathrin Schemann, 2025-03-25T07:30:14.749

Important steps for interpretation and particularly for causal model building

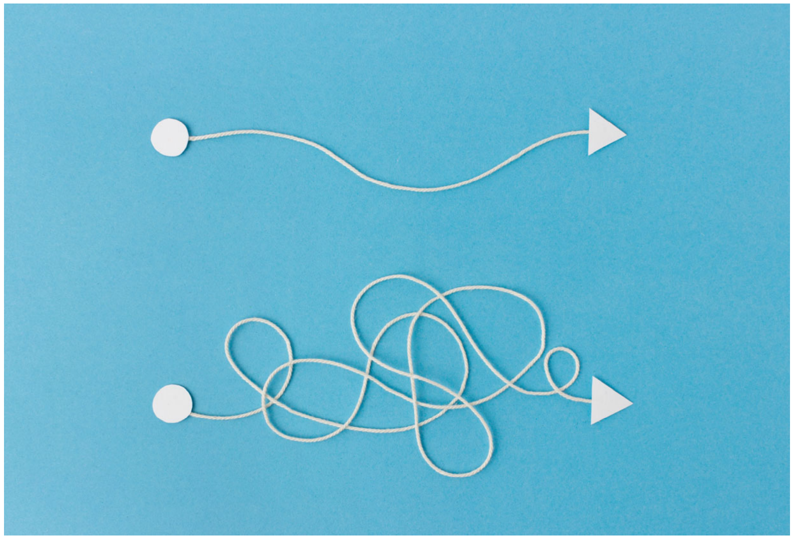
1. Understanding the Question: What specific (causal) relationship are we trying to understand?
2. Exploring the Data: What information do we have about the situation? What factors might be relevant? **Theory**
3. Building a Working Model: Creating a simplified representation of how the outcome might be influenced by different factors (including potential causes). This often involves thinking about how things are connected – **Causal diagram**
4. Considering Potential **Biases**: What hidden factors could be influencing our results?
   - Sampling/ Selection bias
   - Information/ Measurement bias
   - Confounding bias

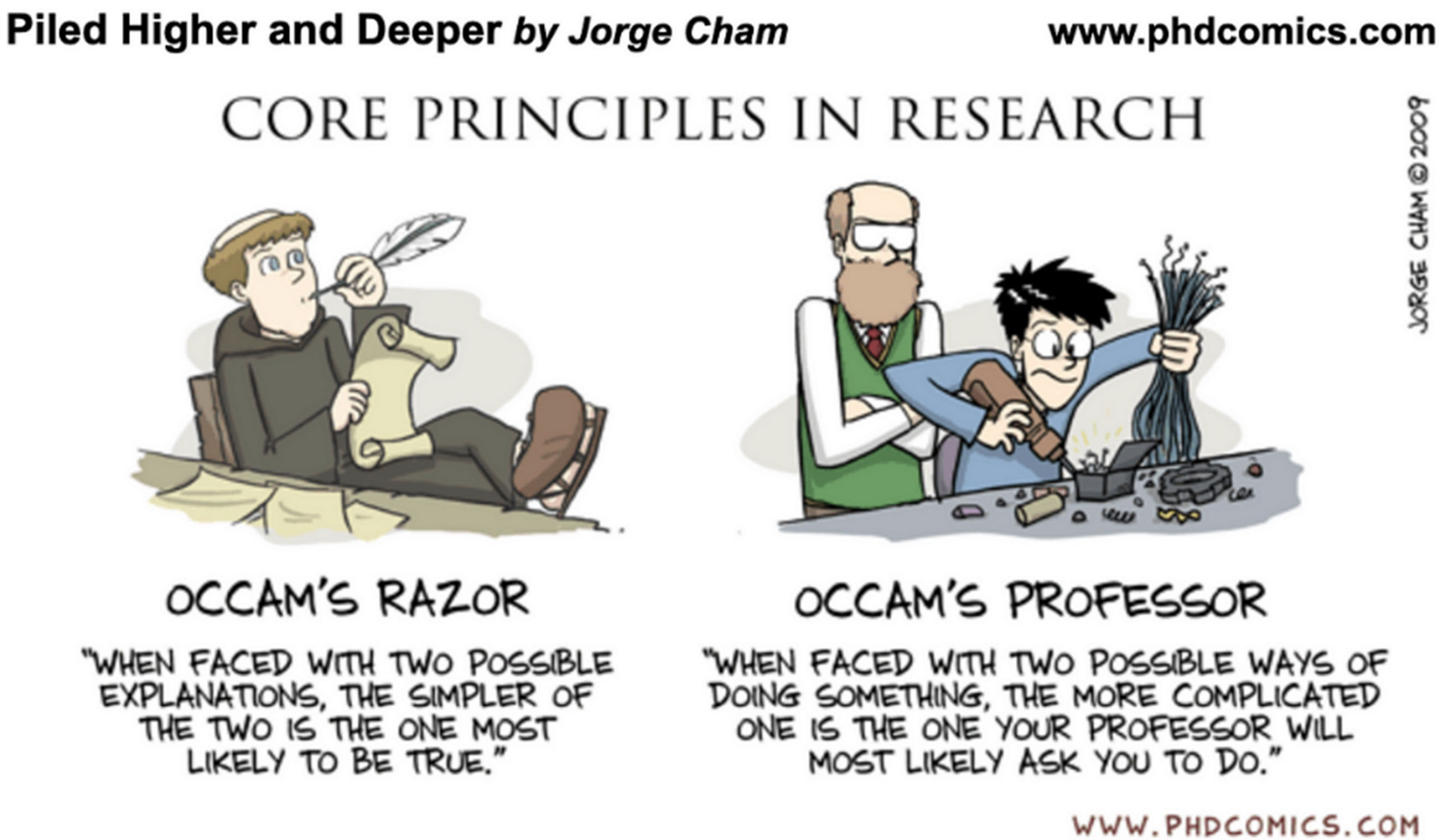
Other Model aim considerations

Subject matter knowledge
- Consider ease of measurement and reliability of variables

Discipline conventions

Parsimony/ Occam's Razor





title: "Core Principles" - originally published 10/12/2009

Experimental and Analytical Design – Model building

Decide on the aim of your model

Create a Statistical Analysis Plan (SAP) and specify your Model Building Strategy.

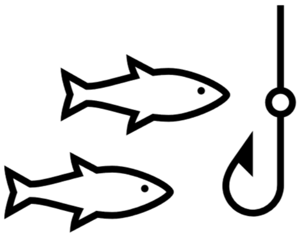
Tip: A sound study design and analysis plan increase your chances of grant success and high impact publications! Consider the SCU @SIH **Research Essentials**, ***Experimental Design*** and ***Power + Sample Size*** *w*orkshops + **SAP template – see our Statistical Resources page.**

Study Design and Analysis Plan - Model Building Strategy

a) Identify the outcome variable and a full set of predictor variables to be considered
b) Pick a suitable modelling method
c) Specify the criterion (criteria) to be used in selecting the variables to be included
d) Specify the strategy for applying the criterion (criteria)

→ Determine potential/known confounders based on theory

→ Determine interactions of interest

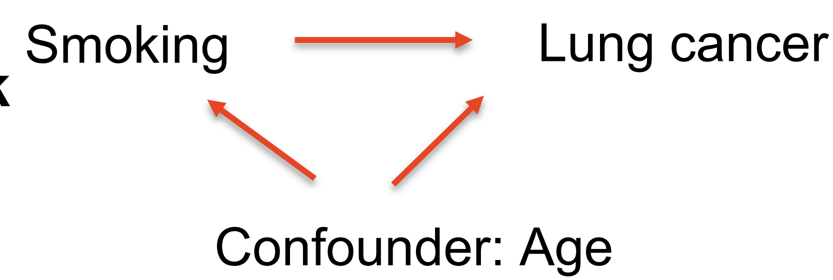


Analysis plan: Identify the outcome variable and a full set of predictor variables to be considered – the full or maximal model

Consider the outcome variable and all possible predictors of interest (justify using literature and unpublished data)

- Consider that administrative data is not designed for/ collected for research purposes
- Include ‘design variables’ - depends on study design and type, controlled versus observational, e.g. block in RCT, exposure of interest, ‘cluster’ variables like students in classes within schools or hospitals or farms and potential confounders – see our ‘*Experimental design*’ workshop for more information
- Consider potential confounders – e.g. age, sex, BMI, SES, comorbidity, previous experience
- Consider what interaction terms to include

This is as much a scientific/clinical task as it is a statistical task.

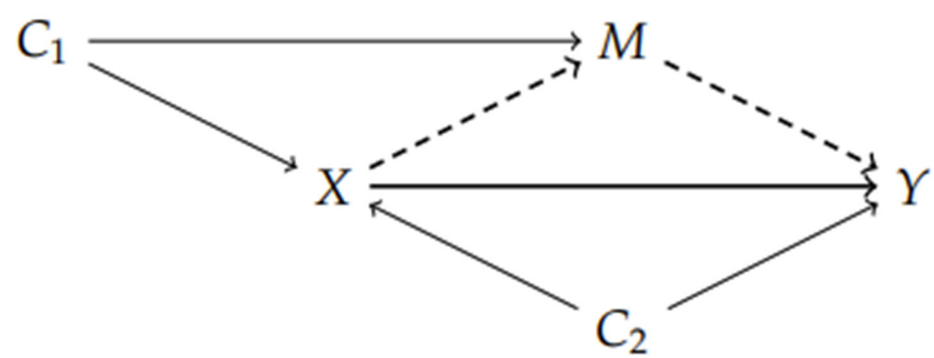


Tip: Documenting the justification for predictor inclusion + relevant references at the design stage will improve your design and make writing the discussion of your paper much easier!

Analysis plan: Identify a full set of predictor variables – other strategies

Causal inference/ strong theory:
If you have strong theory in your area of study, consider the causal structure of your data and data generating process (include sampling and unmeasured variables).

Consider how you can show causation, e.g. using Bradford-Hill criteria; by drawing a causal diagram for predictor variable inclusion/ causal inference (e.g. Directed acyclic graph – DAG, see **Experimental Design** workshop, the appendix and tutorial+ references here)



Propensity Score Analysis

- The Problem with Differences: When we compare groups (e.g., those who received a treatment vs. those who didn't), there might be other differences between them that affect or confound the outcome.

- Propensity Score: A way to account for these differences. It’s the probability of receiving a particular treatment *given* certain characteristics of the individuals.
-
- Think of it as: A score that tells us how likely someone is to get a specific treatment, based on their background.

- By using propensity scores, we can create groups that are more similar in terms of factors other than the treatment itself.

- Inverse probability weighing IPW is the go to method – efficient confounding adjustment (score as predictor takes 1 degree of freedom) but still requires confounding variable identification/selection and does not allow for interpretation of confounders.

Analysis plan: Identify a full set of predictor variables – other strategies

If you have a large number of predictors you may want to:

- Use univariable screening - weak theory/ exploratory:

Start with univariable screening with a liberal p value, e.g. $p<0.2$ or $p<0.25$ and re-test excluded predictors with the final model to ensure you did not miss important confounding; or

- Work in blocks, e.g. all demographic variables first and separately all clinical variables; then combine important predictors in the final model
- For multi-level models consider variables at different levels separately e.g. hospital-level factors versus individual-patient-level factors and enter variables meeting certain criteria from each level cumulative into a final model.

- Consider *inverse propensity score weighing* to adjust for confounders

Analysis plan: Identify a full set of predictor variables - Interactions

Decide what interactions between predictors are to be considered
- Consider multiple testing adjustment if using large number of interactions – see Linear Models 3 workshop
- Evaluate interactions graphically where possible.
- Rule of thumb: Having more than two interaction terms usually over-complicates the model/ interpretation and may not represent reality – so avoid 3-way interactions unless indicated by the subject matter

Five general strategies for creating and evaluating interactions*:

1. Create and evaluate all 2-way interaction terms (ok for number of predictors <=8)
2. Create 2-way interactions among all predictors that are significant in the final main effects model (after model building)
3. Create 2-way interactions among all predictors found to have an unconditional (univariate) association with the outcome.
4. Create 2-way interactions only among pairs of variables which you suspect might interact (based on literature, expert knowledge etc), e.g. those involving the primary predictor of interest and important confounders.
5. Only create 2-way interaction terms that involve the predictor of interest.

***May be adjusted for 'biologically plausible' interactions only.**

Analysis plan: Identify a full set of predictor variables - pitfalls

Over-fitting the data - too many predictors, not enough data/sample size.
Rule of thumb: one should have at least 10 data points for each estimate, better 20
- For logistic regression/ survival analysis one should have at least this number of events

- An intercept needs an estimate/ regression coefficient; Interaction terms need estimates including main effects; categorical variables with more than two groups need multiple.

If the goal is to create knowledge then we need our estimates to be as precise as possible – too little data will lead to large confidence intervals.

A large maximum model should include all potentially important predictors but it increases the chance of multi-collinearity, unstable estimates and finding spurious associations that are not important in the real world or are difficult to interpret. Consider a focused study design collecting high quality data on far fewer predictors versus administrative ‘big’ data not collected for research purposes.

Take home: Consider sample size as part of your study design/analysis plan to ensure your model is sufficiently powered! See our ***Power and Sample Size*** workshop for more information.

Analysis plan - Pick a suitable modelling method

Outcome variable:
Its data type indicates which types of models may be used

See our specific workshops for more information on different models, e.g.:

- Linear Models 1 (numeric outcome)
- Linear Models 2 (binary outcome or count data)
- Survival analysis (time-to-event data).

Analysis plan: Specify the selection criteria

How to decide which variables to keep in the model?

Selection criteria are formal 'Goodness of Fit' criteria and other statistical considerations

Other statistical considerations - retain variables that are:

- a primary predictor of interest
- a priori confounders for the primary predictor of interest
- Shown confounders for the primary predictor of interest (should not be an intervening variable – see appendix)
- A component of an interaction term included in the model (for interpretation)

Analysis plan: Specify the selection criteria

Formal Goodness of Fit criteria – nested models
Nested models are based on the same set of observations (n) and the predictors in one model are a subset of predictors in the other model

Tests based on nested models to evaluate significance of a predictor:

- Partial F test (linear model)
- Wald test or likelihood ratio test (LRT) (other types of regression such as logistic, Poisson) – Wald test often most convenient, but LRT has best statistical properties and should be used if p value or standard error are questionable

Formal Goodness of Fit criteria– non-nested models
Information Criteria: IC = -2 log-likelihood + alpha*number of parameters

Akaike's Information Criteria (AIC): alpha = 2
Bayesian Information Criteria (BIC) alpha = log n (with some variation)

Analysis plan: Specify the selection criteria

AIC/BIC:

- Assess overall model so can be used to compare different models
- The smaller the IC, the better the model
- Can be used to compare nested and non-nested models
- Can be used to compare different regression models, e.g. linear vs Poisson
- Can't be used to compare models based on different sets of observations
- Can't be used to compare models with different likelihood computation, e.g. Cox semi-parametric survival model versus Weibull parametric model
- BIC depends on n and it can be unclear what n to use for clustered data
- BIC favours most parsimonious model

Absolute difference in AIC	Absolute difference in BIC	Evidence for superiority of the better model
0- <4	0 - <2	Weak
4- <7	2- <6	Positive
7 - <10	6 - <10	Strong
=>10	=> 10	Very strong

Analysis plan: Specify the selection criteria

Two additional approaches for linear regression models:

R squared - amount of variance explained by a univariable model
Adjusted R squared = amount of variance explained by a multi-variable model, penalised for model complexity/number of predictors (larger is better)

- Avoids including predictors that explain a small amount of variance only
- Maximising R squared, minimises mean square error (MSE)

Mallow's Cp Statistic (special case of AIC) :

Cp = Sum((Y-Y_hat)^2)/sigma^2) – n + 2k

p = number of parameters in the model (including intercept)

Situation	Meaning
$C_p \approx p$	Good model (low bias, good fit)
$C_p > p$	Model may be missing important variables (biased)
$C_p < p$	Possible overfitting or variance issues

Analysis plan: Specifying the selection strategy

How to get from a full/maximal model (all predictors selected for model building) to the final model?

→Choose a selection strategy to build a model

- Manual options
- Automated processes (be very cautious!!)
- Other considerations

Analysis plan: Specifying the selection processes – the options

All possible:
Examine all possible combinations of predictors.

Best subset:
Software identifies ‘best’ model of all possible based on criteria (e.g. a model with 2 predictors, largest adjusted R squared; a model with 3 predictors, largest adjusted R squared)

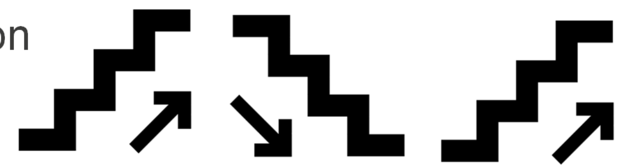
Forward selection
First a model with only the intercept is fitted and then each variables is added selectively based on a specified criterion, e.g. having the largest Wald test statistic provided it corresponds to $p<0.05$ (or other chosen significance level). The predictor with the largest Wald test statistic is added first and then the process is repeated and continues until no term meets the entry criterion.

Backward elimination
As for forward selection but the process is reversed and model building starts with the full/maximal model. Predictors are removed sequentially until none of the predictors remaining in the model has a Wald test statistic meeting the specified criterion.

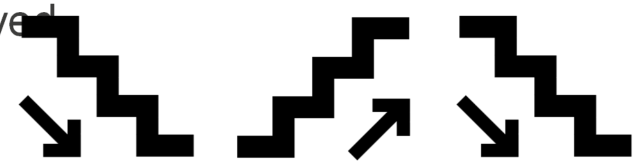
Analysis plan: Specifying the selection process – the options

Stepwise regression
A combination of forward selection and backward elimination.

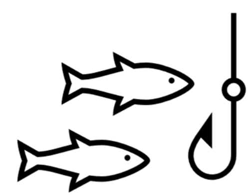
Forward stepwise starts with forward selection but after the addition of each variable, the criterion for backward elimination is applied to each variable in the model to see if it should remain.



Backward stepwise starts with a full/maximal model and sequentially removes predictors but after the removal of each variable, all removed variables are checked to see if any of them would meet the forward selection criteria for inclusion.



Stepwise methods are critisised by some to be data dredging/ fishing approaches.



So, what strategy to choose?

Analysis plan: Variable selection process - comparison

Selection strategy	Comment
All possible	Good for early exploratory work with few predictors to find multiple good model candidates. Can compare nested and non-nested models.
Best subset	Researcher identifies when increasing the number of predictors brings little predictive improvement. Can compare nested and non-nested models.
Forward	Supports start simple, get complex – understanding at each step. Can assess a priori confounders. May miss an important confounder
Backwards	Statistical significance of terms is assessed after adjustment for potential confounders; useful for smaller number of variables e.g. several demographic that are likely confounding.
Forward stepwise	Useful for large number of predictors/interaction terms.
Backwards stepwise	Generally favoured over forward stepwise.

Analysis plan: Specifying the selection strategy

Automated approaches:
Be cautious! While convenient, they should be considered exploratory methods rather than definite approaches.
- some journals will no longer accept automated selection
- They yield R squared values that are too high
- Based on Hypotheses tests – p values too small and not adjusted for multiple testing → see LM3 Workshop
- Ignore multi-collinearity – important to do your EDA/ correlation analysis!! Use VIF after model building to check.

- DO NOT incorporate other statistical criteria:

Select 'design' variables/predictor *a priori*, e.g.
  - Predictor variable of interest for the research question
  - Randomisation blocking factor in an experiment
  - *A priori* confounder of the predictor variable of interest

Model building strategy styles (adapted from Keith McCormick)

	Hierarchical	Simultaneous	Stepwise
Style	most academic		least academic
Theory	Strong theory	Limited theory	no theory
Analyst role in model building	choose variables, and the order of entry	choose a list of variables believed to be important	Variables are chosen through automated process
Possible use	Designed experiments	Exploratory	Data mining type approach

Source: LinkedIn Learning video: Three regression strategies, Keith McCormick.

Analysis plan: Model Building Strategy - Collett

Collett's general strategy for model selection:

1. Fit univariate models for each predictor variable of interest. Compare the -2loglikelihood values to the null model (using chi square statistic, or AIC, BIC)
2. Significant variables from step 1 are fitted in a single model and compared to 'leave one out' models
3. Variables omitted at step 1 are then added to the best model from step 2 and compared
4. A final check to ensure no term in the model can be omitted without increasing -2LL significantly and no new term added without reducing -2LL significantly.

Pitfalls with automated selection processes:

Avoids researcher thinking about their data and taking responsibility for their analysis!

“Start Simple – Get Complex”

“Every model you run tells you a story. Stop and listen to it.”

- Avoid Overfitting (have at least 10 (better 20!) observations per parameter in the model – keep in mind dummy coded categorical variables with > 2 categories e.g. a predictor with 5 categories requires 4 parameters to be estimated)
- For interpretability of parameters keep the main effects of both variables that make up a significant 2-way interaction term in the model (irrespective of their p values)
- Analysis will only be based on observations for which all variables are not missing (with many missing observations the final data set analysed may be a small subset) – do EDA!

Workflow: Steps in Model Building so far

1. Experimental and Analytical Design
    - Decide on the aim of your model
    - Create an analysis plan and specify your Model Building Strategy

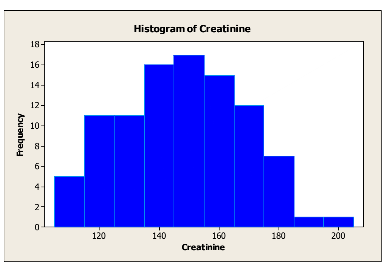
2. Data Cleaning

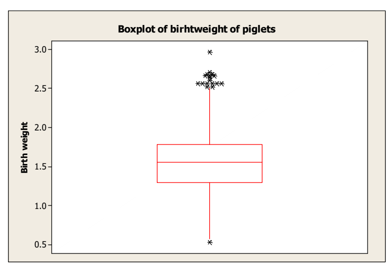
Step 2: Data Cleaning

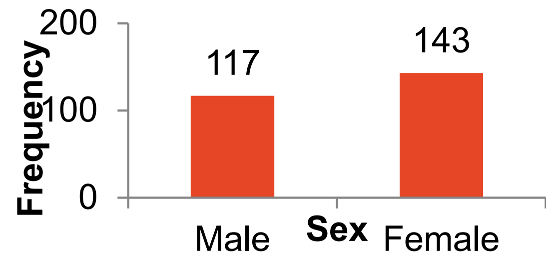
For each variable including the outcome: identify the data type (numeric/categorical) and use appropriate summary statistics and plotting to check the distribution:

- Numeric variable – histogram/boxplot; mean, median, standard deviation, percentiles, etc.
- Categorical variable – bar charts; frequency tables with count and percent
- For model building we want variables that:
  - are measured accurately, precisely + are reasonably complete (not too much missing data, e.g. no more than 10-15% missing observations)
  - have substantial variability (e.g. if 99% are male, than sex is not a good predictor)

  Categorical variables: consider combining categories with small number of observations/ eliminate.







➔ See our Research Essentials – Analysing your Data Workshop for further details

Workflow: Steps in Model Building so far

1. Experimental and Analytical Design
   - Decide on the aim of your model
   - Create an analysis plan and specify your Model Building Strategy
2. Data Cleaning
3. Exploratory Data Analysis (EDA): Pick predictors to fit and a suitable model using EDA

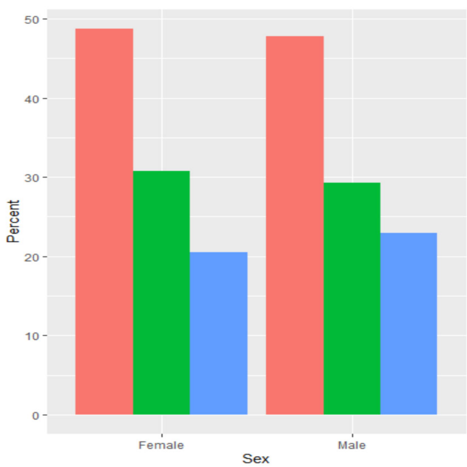
   Assess relationships between each predictor and the outcome

   Assess the relationships among predictors and consider variable reduction

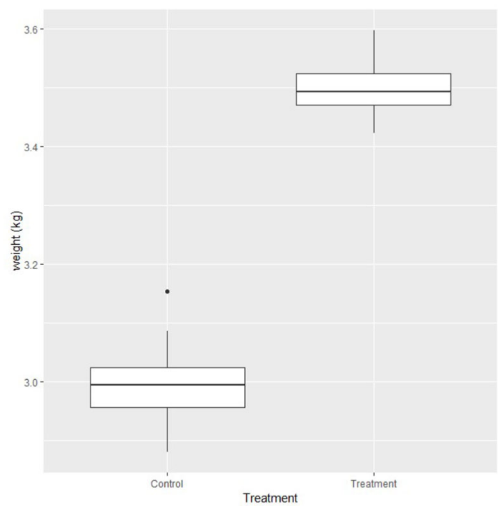
Step 3: Pick predictors to fit using Exploratory Data Analysis (EDA)

3.1 Assess relationships between each predictor and the outcome to select, modify + understand predictors

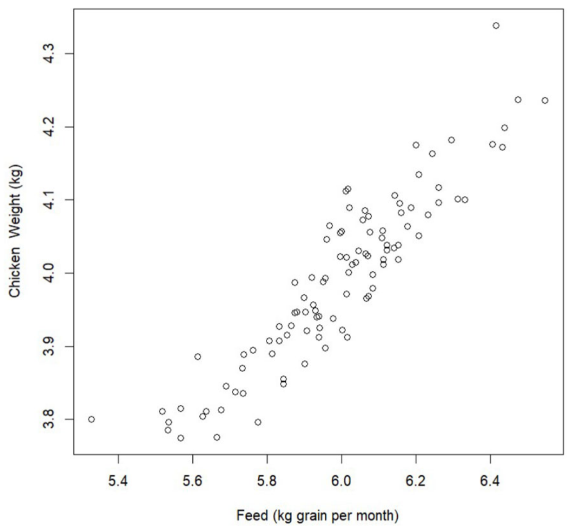
Plot the relationship of each predictor with the outcome – see ***Research Essentials***



2 categorical variables
- side-by-side bar charts



1 categorical, 1 numeric variable
- side-by-side boxplots



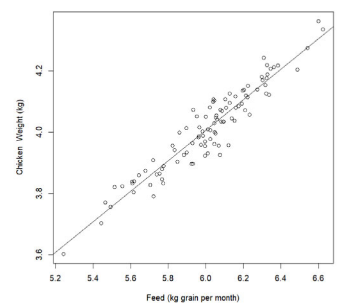
2 numeric variables
– xy scatter plot

Step 3: Pick predictors to fit using Exploratory Data Analysis (EDA)

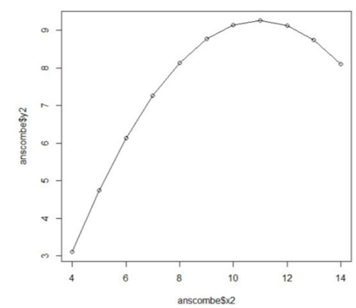
3.1 Assess relationships between each predictor and the outcome to select, modify + understand predictors

For numeric variables in linear regression – assess the assumption of linearity – is the relationship between outcome and predictor a line, a curve or something else?

For practical reasons we assess the model assumption of linearity before model building – see *Linear Models* workshops.



Linear Model (a straight line):
$y = b_0 + b_1 x_1 + error$



Quadratic model (a curve):
$y = b_0 + b_1 x_1 + b_2 x_1^2 + error$

Alternatively, avoid the assumption by categorising the numeric predictor, but this loses information and may introduce bias. However, sometimes we are interested in specific categories for interpretation, e.g. BMI – underweight/normal weight/ overweight/obese.

Step 3: Pick predictors to fit and a suitable model using Exploratory Data Analysis (EDA)

3.1 Assess relationships between each predictor and the outcome

3.2 Assess the relationships among predictors and consider variable reduction

WHY is EDA so important??

"Knowledge without practice is useless; practice without knowledge is dangerous."
Confucius

A case study…

Consider results from a multivariable model for Math test score…

	Estimate	Std. Error	t value	P value
(Intercept)	2.105	1.104	1.9	0.0569
IQ	0.997	0.006	151.9	<0.001***
verbal.IQ	-9.995	0.067	-148.3	< 0.001***

Multiple R-squared: 96%

$$y = b_0 + b_1 x_1 + b_2 x_2$$

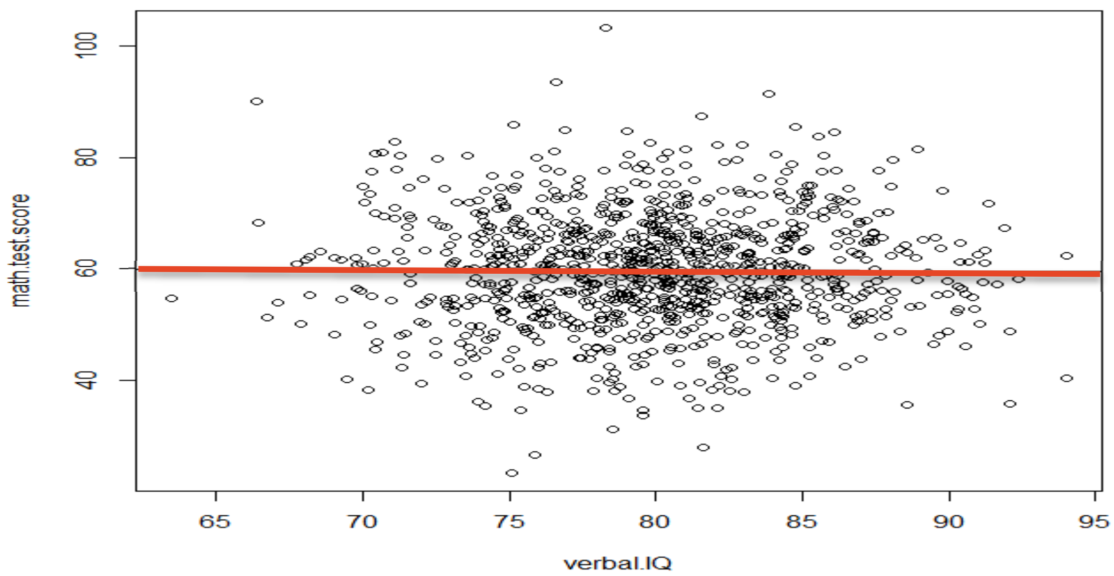
Math test score = 2.1 + 1*IQ – 10*verbal.IQ

What is your interpretation? Is this a good model?

EDA for verbal IQ and Math test score:

EDA: correlation r=0 → there is no relationship!

Multivariable Model: on average with a 1 point increase in verbal IQ, Math test score decreases by 10 points!

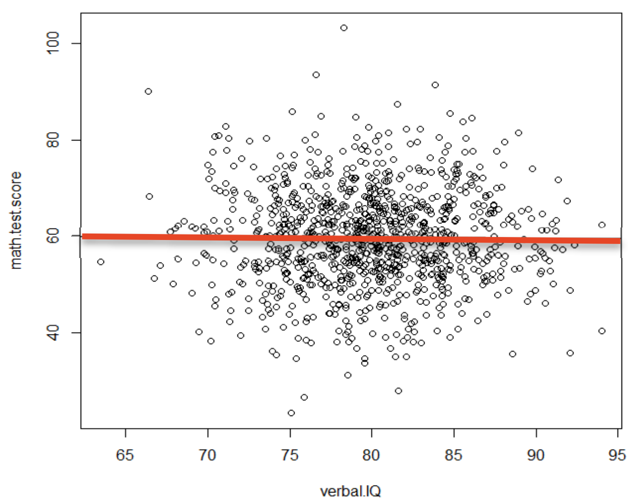


Start Simple – Get Complex: use Exploratory Data Analysis (EDA) to assess the relationships among predictors and consider variable reduction

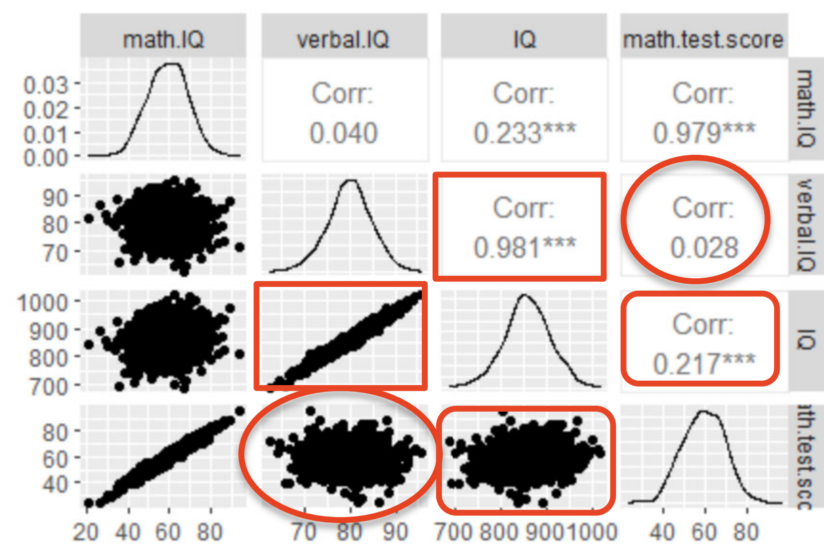
Numeric predictors: use xy scatter plot and conduct **pairwise correlation analysis**

‘High’ correlation is domain-specific, generally r>|0.7|,|0.8|or |0.9| but r<0.7 can be problematic too

Only use one of a pair of highly correlated variables for multivariable modelling – **WHY?**



Correlation coefficient r = 0



Correlation matrix

Simulated data

```
math.IQ <- rnorm(1000, 60, 10)
math.test.score <- math.IQ + rnorm(length(math.IQ),0,2)

verbal.IQ <- rnorm(1000, 80, 5)

IQ <- 10*verbal.IQ + math.IQ
```

- math.test.score is math.IQ with a bit of noise
- IQ is a sum of verbal IQ and math.IQ (IQ:verbal.IQ r = 0.98)
- A model with verbal.IQ and IQ can re-arrange itself to:

Math.IQ = IQ – verbal.IQ

Example summary

Results:

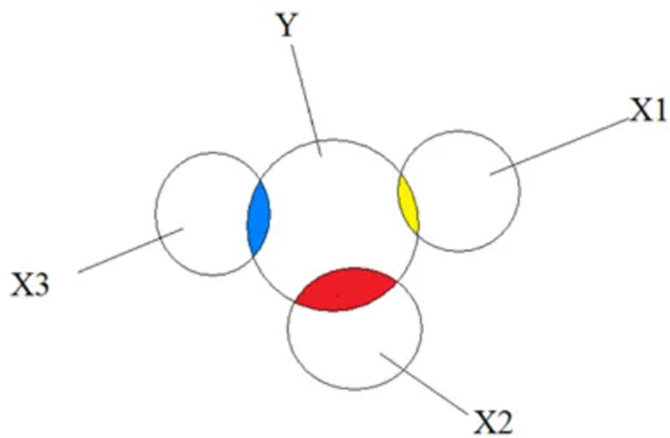
- *Verbal IQ* is not negatively correlated with *Math Test Score* as shown by the simple marginal model and the simulated data
- Yet a model with *Verbal IQ* and *IQ* suggests *Verbal IQ* is negatively corelated with *Math Test Score*.
- WHY: because *IQ=verbal IQ* + *math IQ meaning* a model which has both *IQ* and *verbal IQ* (which are correlated) can rearrange itself as *math IQ = IQ – verbal IQ*.
- This a reversal of *verbal IQ - verbal IQ* is SUPRESSING *IQ* (the part of IQ that is uncorrelated with Math results).

➔This is an example of the reversal paradox due to multi-collinearity – this may be called *Simpson's paradox*, *Lord's paradox*, and *suppression* - depending on whether the outcome and explanatory variables are categorical, continuous or a combination of both

➔This is why we always look for multi-collinearity during the Exploratory Data Analysis using a scatterplot matrix to learn about the relationships between variables.

➔Start Simple – Get Complex!

Relationships between variables - Correlation

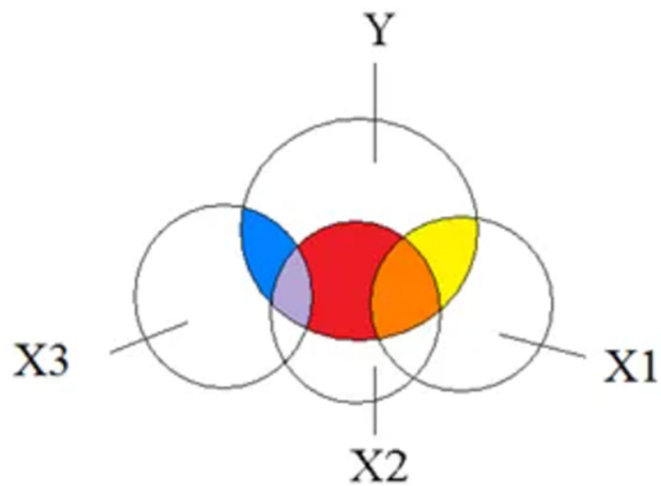


Why is 'high' correlation an issue for model building?
When interpreting the model, we make an assumption that the predictors are 'independent'.

What is correlation? This is NOT it!! X1, x2 and x3 are statistically independent. No conditional effects – effects the same as in univariate analysis.

correlation r=0

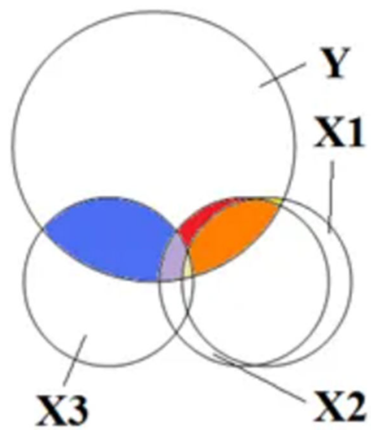
Relationships between variables - Correlation



Moderate correlation: there is overlap, but conditional effects are still interpretable.

correlation r=0.5

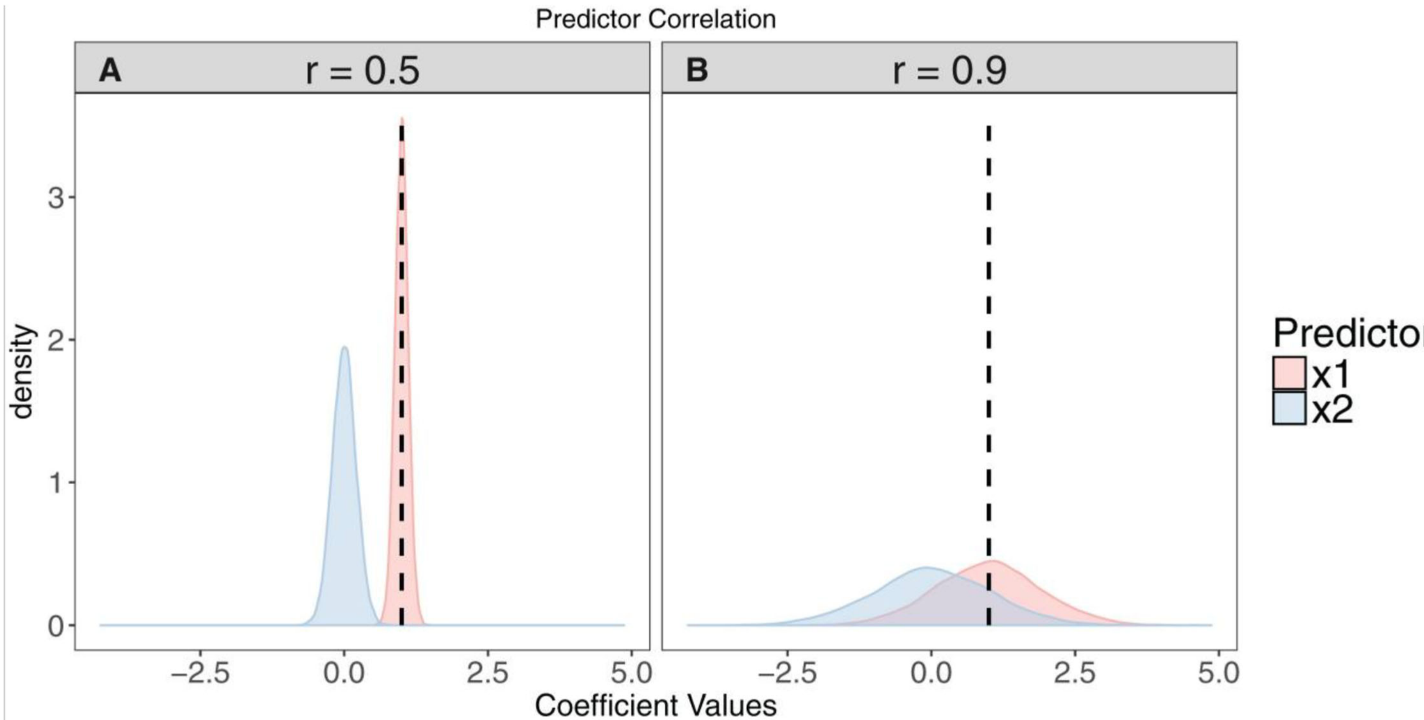
Relationships between variables - Correlation



High correlation: x1 and x2 are almost identical

correlation r=0.9

The effect of high correlation: Example: y ~ x1 + x2



Estimate + 95% CI

Moderate correlation **High correlation**

Effect of high correlation/multi-collinearity on modelling

Extent of Multicollinearity	Effect on the Regression Analysis
Little	Not a problem
Moderate	Not usually a problem
Strong	Statistical consequences: Often a problem if you want to estimate effects of individual *X* variables (ie, regression coefficients); may not be a problem if your goal is just to predict or forecast *Y*
Extremely strong	Numerical consequences: Always a problem; computer calculations may even be wrong due to numerical instability

Potential indicators of multi-collinearity:

- Coefficients have signs opposite to what you'd expect from theory (suppression – see DAG in appendix for explanation)
- Very high standard errors for regression coefficients
- Overall model is significant, but none of the coefficients are
- Large changes in coefficients when adding predictors
- Coefficients on different samples are wildly different
- High Variance Inflation Factor (VIF) and low tolerance (VIF reciprocal) – direct measure of how much the variance of the coefficient is being inflated due to multicollinearity – e.g. linear combinations of variables correlated with a variable
- High condition index in PCA – ratio between first and last PC

For further information see:
https://www.theanalysisfactor.com/eight-ways-to-detect-multicollinearity/

Step 3: Pick predictors to fit using Exploratory Data Analysis (EDA)

3.2 Assess the relationships among predictors and consider variable reduction

High pairwise correlations – what to do?

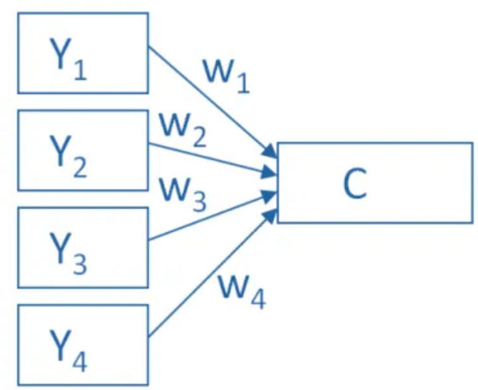
- If few variables: select only one of the pair based on biological plausibility, fewer missing values, ease/reliability of measurement or lower univariate p value (may not fix problems arising from collinearity among linear combinations of predictors --> also check Variance Inflation Factors (VIF) after analysis)
- If many (related?) predictors: explore their relationship/consider reducing the number of variables by using multivariate techniques - see our Multivariate Statistical Analysis 1 workshop
- Summarise the variables by creating an index/scale

-

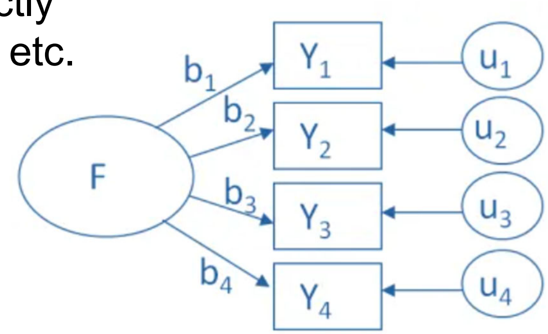
Multivariate analysis - Principal component Analysis + Factor Analysis

Principal component Analysis (PCA) and Factor Analysis (FA) are data reduction techniques to consolidate the information contained in a set of numeric predictor variables into a new set of fewer, uncorrelated variables. If used subsequently in a model one cannot statistically test individual predictors.

PCA:
Creates one or more index variables (components) from a larger set of variables by using linear combinations (basically a weighted average) of a set of variables.
Component coefficients from a subsequent model can be back-transformed into coefficients for the original predictors – these are more stable as multi-collinearity is avoided.



FA:
-Models the measurement of a latent variable which cannot be directly measured with a single variable, e.g. intelligence, statistical anxiety etc.
-For subsequent modelling, determining which original predictor is important is subjective based on high correlations/factor loadings.

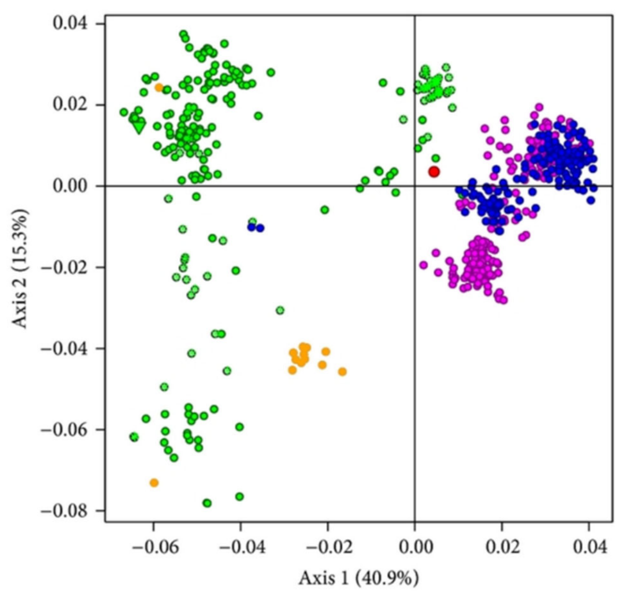


See SIH Multivariate Statistical analysis 1 workshop.

Multivariate analysis - Correspondence analysis

- A form of exploratory data analysis (EDA) designed to analyse the relationships among a set of categorical predictor and outcome variables
- Produces a visual summary which is a scatter plot with factorial axes that reflect the most variability in the original predictor variables
- Allows identification of clusters of predictors that are closely associated, with clusters further from the intersection of the axes having stronger associations

Summary:
PCA/FA and correspondence analysis are complementary techniques to modelling and provide insight into relationships between groups of predictors and there association with the outcome



Create an index or scale

Combine a number of related predictor variables into a single index

- Subjectively, ideally based on prior research - can have different weights for different contributing factors, e.g. an infection control index may be created by counting the number of hygiene practices performed and expressing this as a percentage of all practices. This could also be categorised into low/medium/high.
- Objectively, e.g. fan capacity, size and number of air inlets and building size may be used to calculate the number of air changes per hour which could also be expressed as proportion of recommended ventilation level
- When predictors are assumed to be reflective of an underlying, unmeasured characteristic (a latent variable) can combine into index or scale by summing or averaging predictors and use Cronbach's alpha statistic to evaluate internal consistency of the scale. Use correlation analysis to assess correlation between each item (variable) and the scale and pairs of items to identify any items that do not fit into the scale well.
    - See SIH workshop on **Surveys' 2** for further information

Create an index or scale - Comorbidity

Definition:

Comorbidity is defined as the co-occurrence of one or more disorders in the same patient either at the same time or in some causal sequence and a common confounder

- Indices are calculated (weighted sum) from international ICD-10 diagnoses codes (+/- other data) from administrative data (e.g. admitted patients data)

- Indices are based on Elixhauser, the Charlson/Deyo, and the Charlson/Romano methods and well-established for risk adjustment and mortality prediction

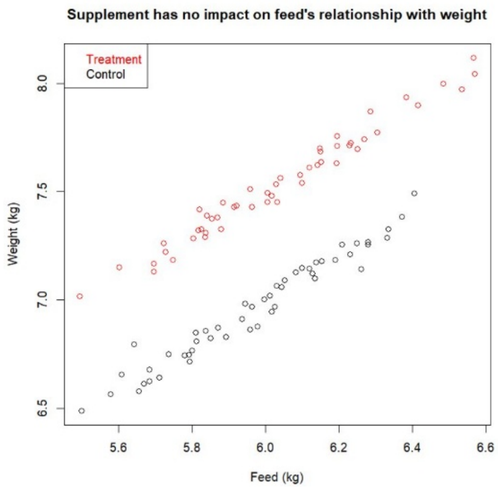
- Calculation in R software: comorbidity package – Rdocumentation

Charlson comorbidity index

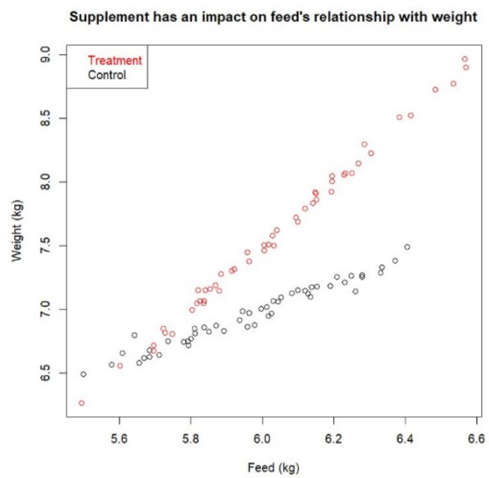
Condition	Points in CCI
Myocardial Infarction	1
CHF	1
Peripheral Vascular Disease	1
Cerebrovascular Disease	1
COPD	1
Dementia	1
Paralysis	1
Diabetes	1
Diabetes With Sequelae	2
Chronic Renal Failure	2
Various Cirrhodites	1
Moderate-Severe Liver Disease	3
Ulcers	1
Rheumatitis	1
AIDS	6
Any Malignancy	2
Metastatic Solid Tumor	6

Step 3: Pick a suitable model and predictors to fit using EDA

3.2 Plot the data in relation to the research question, e.g.
Plot multiple variables in one plot; plot variables over time



Parallel lines: consistent effect



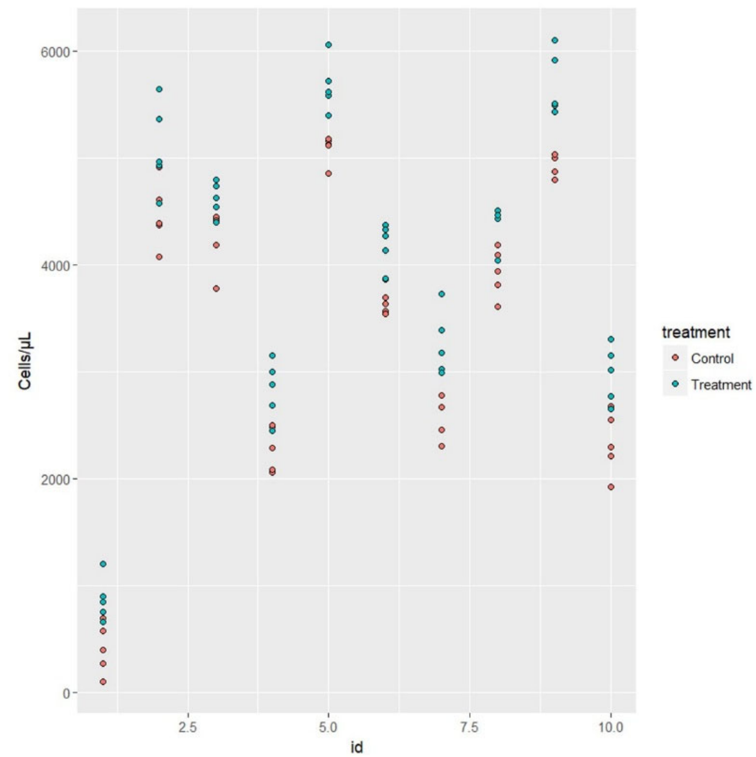
Non parallel lines: inconsistent effect → add interaction

Step 3: Pick a suitable model and predictors to fit using EDA

Plot data for individuals:

The difference between the 10 people (id) is much bigger than the effect of treatment.

→ To account for the variation between people we use a random effect for person in a mixed model – this makes our model more accurate and the estimate of treatment effect more precise.



Step 3: Pick predictors to fit using Exploratory Data Analysis (EDA) - Summary

EDA for selection/modification of predictor variables for model building:

3.1 Assess relationships between each predictor and the outcome

- Plot relationships and consider excluding variables with low variability
- Assess assumption of linearity for numeric predictors- see *Linear Models* workshops

3.2 Assess the relationships among predictors and consider variable reduction

- Check for multi-collinearity and only select unrelated variables into model building
- Consider using univariable screening to reduce the number of variables for model building
- Look for patterns/structure in your data, e.g. interaction/ repeated measures

R code/ resources for data cleaning and EDA

R code: EDA
- Introductory R and visualisation for EDA: R-essential-training-wrangling-and-visualizing-data
- Data management and summaries with R tidyverse: Learning the R Tidyverse
(These two LinkedIn Learning course are free with your Sydney uni login details)

R correlation matrix example:

```
# Create/simulate data
math.IQ <- rnorm(1000, 60, 10)
verbal.IQ <- rnorm(1000, 80, 5)
IQ <- 10*verbal.IQ + math.IQ
math.test.score <- math.IQ + rnorm(length(math.IQ),0,2)
my_data <- data.frame(math.IQ, verbal.IQ, IQ, math.test.score)

# install Ggally package and run ggpairs on all numeric variables (in this example all four variables are numeric)
install.packages("GGally")
library(GGally)
ggpairs(my_data)
```

Workflow: Steps in Model Building so far

1. Experimental and Analytical Design
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Data/Inferential Analysis for Interpretable Models

Fit a single model LM1/2/3, Survival analysis

1. Check model assumptions
2. Check goodness-of-fit
3. Interpreting Model Parameters and reach a conclusion

Iterative Model building

- Pick predictors to fit and a suitable modelling method based on EDA
- Iteratively fit and assess models using predetermined strategy

Step 4: Data/Inferential Analysis – Model Building

a) Use iterative model building and determine final multivariable base model + check assumptions and goodness-of-fit
b) Check interactions of interest with final base model if this strategy was pre-specified
c) If screening used: reassess eliminated predictors by adding one-by-one to final model
d) Assess models: enumerate level of confounding – calculate % change in estimate
e) Interpret model building strategy: determine final, final model – consider sensitivity analysis (e.g. model with and without potential confounder when $p>0.05$)
f) Prediction – rank models in order of predictive ability

Step 4: Inferential Model building

- An iterative process
- Fit and evaluate models based on predefined criteria and strategy

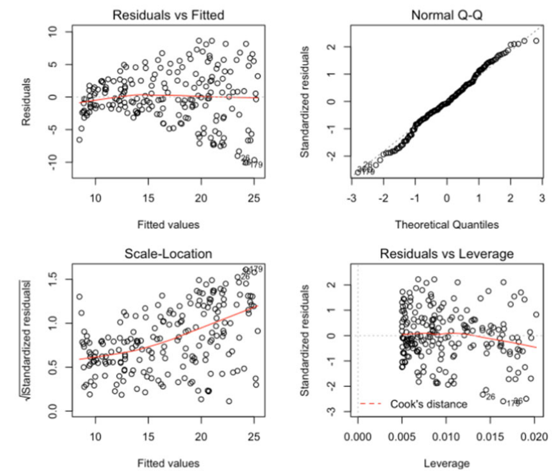
Consider changing strategy if:
- issues with sampling/ optimal sample size not reached
- EDA indicates modelling method not suitable

Variables in the Equation

		B	SE	Wald	df	Sig.	Exp(B)	95.0% CI for Exp(B)	
								Lower	Upper
Step 1	Age at hospital admission	.066	.006	118.799	1	.000	1.068	1.056	1.081
Step 2	Age at hospital admission	.059	.006	92.407	1	.000	1.060	1.048	1.073
	Congestive heart complications	-.861	.142	36.570	1	.000	.423	.320	.559
Step 3	Age at hospital admission	.059	.006	92.280	1	.000	1.061	1.048	1.074
	Cardiogenic shock	-.883	.261	11.414	1	.001	.413	.248	.690
	Congestive heart complications	-.820	.143	32.745	1	.000	.440	.332	.583
Step 4	Age at hospital admission	.060	.006	90.699	1	.000	1.061	1.048	1.074
	hr	.009	.003	10.649	1	.001	1.009	1.004	1.015
	Cardiogenic shock	-.959	.261	13.464	1	.000	.383	.230	.640
	Congestive heart complications	-.707	.147	23.109	1	.000	.493	.370	.658
Step 5	Age at hospital admission	.054	.006	71.465	1	.000	1.055	1.042	1.069
	hr	.012	.003	16.723	1	.000	1.012	1.006	1.018
	initial diastolic BP	-.012	.003	11.022	1	.001	.989	.982	.995
	Cardiogenic shock	-1.052	.264	15.875	1	.000	.349	.208	.586
	Congestive heart complications	-.690	.147	22.087	1	.000	.501	.376	.669
Step 6	Age at hospital admission	.048	.007	52.485	1	.000	1.049	1.036	1.063
	hr	.012	.003	16.746	1	.000	1.012	1.006	1.018
	initial diastolic BP	-.011	.004	9.952	1	.002	.989	.982	.996
	BMI	-.046	.016	8.059	1	.005	.955	.926	.986
	Cardiogenic shock	-1.099	.265	17.199	1	.000	.333	.198	.560
	Congestive heart complications	-.692	.146	22.319	1	.000	.501	.376	.667
Step 7	Gender	.294	.144	4.190	1	.041	1.341	1.013	1.777
	Age at hospital admission	.050	.007	56.392	1	.000	1.051	1.038	1.065
	hr	.012	.003	18.357	1	.000	1.013	1.007	1.018
	initial diastolic BP	-.012	.003	11.281	1	.001	.988	.982	.995
	BMI	-.048	.016	8.530	1	.003	.953	.923	.984

Step 4: Inferential analysis - Check the model assumptions

Use “diagnostics” to assess the *validity* of the model – see *Linear Models* and *Survival Analysis* workshops e.g.
- Evaluate normality of residuals in LM
- Check HR/OR proportionality assumptions
- Check for influential values

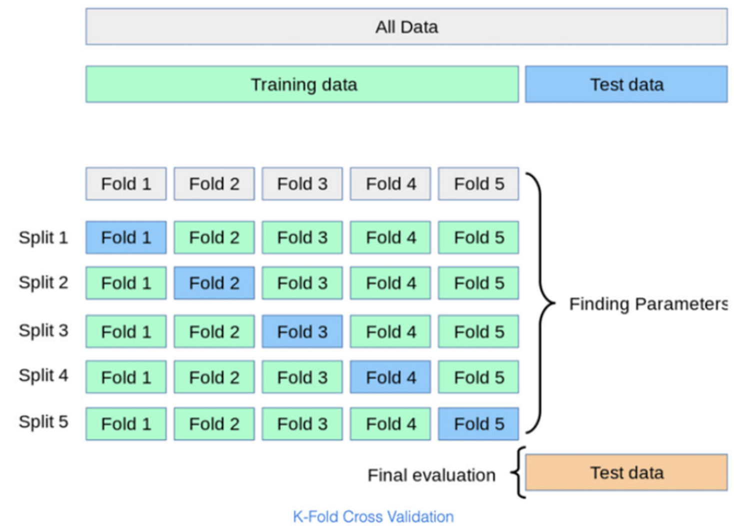


For multivariable models:
Check the assumption of independent predictors using Variance Inflation Factors (VIF) for multi-collinearity among more than 2 predictors after analysis:
VIF >4 moderate multi-collinearity
VIF >10 strong multi-collinearity

Check model fit

If model is predictive: Evaluate the *reliability* of the model – how well will the model predict observations in future samples?

Different approaches, e.g.:
- Split-sample analysis
- Cross-validation
- Leave-one-out analysis
- Bootstrap



- Goodness-of-fit test, e.g. Hosmer-Lemeshow test

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Iterative Model building

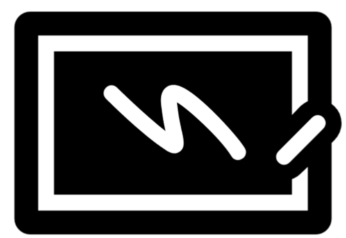
- Pick predictors to fit and a suitable modelling method based on EDA
- Iteratively fit and assess models using predetermined strategy

5. Publication/Report

Step 5: Publication/reporting – interpret results

- Present point estimates – the coefficients (including the intercept)
- Present the measure of uncertainty – their standard errors and/or their confidence interval.

Depending on target audience, non-numerical presentation of study results may be preferable.



Step 5: Interpret and report the results

Assessing the impact of continuous variables:

- Continuous variables are measured on different scales, so a ‘one unit change’ could be small or large.
- It is difficult to compare the impact of different continuous variables

→ use standardised coefficients (multiply by ratio of SD of predictor to SD of outcome)

→ compute and present a range of predicted effects as a continuous predictor changes over its IQR

Variable	Estimate	Basis	Estimated effect change	Effect
Held-back	0.666	dichotomous	O - 1	0.666
Herd size	0.669	IQR	0.55-1.60	0.702
experience	0.023	IQR	8.5 – 26.0	0.401

Modelling log-prevalence of respiratory disease in pigs

Step 5: Interpret and report the results

Control of confounding at the analysis stage:

Use change in measure of association as an indication of confounding

Calculate % change between estimates for statistically significant predictor of interest with confounder in the model versus without confounder in the model

Include confounder as covariate in regression analysis; assess confounding effect by examining the % change in regression parameter estimate in a model with and without a (potential) confounder (see next slide for an example calculation).

Confounder are often included *a priori*, irrespective of p-value or the effect of its inclusion/exclusion on change in other parameters.

Assessment of confounding should be specified in your analysis plan and a “substantial difference” defined *a priori*, e.g. >20-30% change in log odds may be considered substantial confounding.

Remember: Don’t forget to describe your treatment of confounders in the “Methods” section of your paper and report the results of your assessment in the “Results” section.

Step 5: Interpret and report the results – example results tables

Table 1: Univariable linear regression model results showing the association between offspring height with father's height, mother's height, gender and financial situation in a study of 288 university students conducted in Sydney.

Variable		$\hat{b}$	S.E.($\hat{b}$)	95% CI	P value	Percentage variance accounted for (Adjusted R^2)
Constant		90.0	12.3	(65.9, 114.1)	<.001	13.7
Father height		0.47	0.07	(0.33, 0.60)		
Constant		71.4	14.3	(43.2, 99.6)	<.001	15.1
Mother height		0.62	0.09	(0.45, 0.79)		
Gender (reference=Female)		167.0	0.56	(165.9, 168.1)	<.001	46.8
	Male	13.98	0.88	(12.2, 15.7)		
Finance (reference=Finding it difficult)		171.5	1.17	(169.2, 173.8)	0.147	0.8
	Just getting along	0.33	1.69	(-3.00, 3.66)		
	Comfortable	0.61	1.59	(-2.52, 3.74)		
	Prosperous	3.63	1.73	(0.23, 7.03)		

Going from a univariable model to a multivariable model with gender and father height, the effect of mother height decreases by 33%.
= ((0.62-0.41)/0.62)*100

Table 2: Final multivariable linear regression model for offspring height in a study of 288 university students in Sydney

Variable		Estimate	S.E.	95% CI		P value
Intercept		24.70	10.20	4.71	44.69	0.016
Mother_ht		0.41	0.05	0.31	0.52	<.001
Father_ht		0.42	0.04	0.34	0.51	<.001
Gender	Male versus Female	14.19	0.69	12.84	15.53	<.001

Step 5: Interpret and report the results

Predictors eliminated from a model – different options, particularly for exploratory observational studies:

- Eliminated predictors may be forced back into the final model one-by-one to check its effect – if this method is used it should be specified in the analysis plan + paper methods section
- You may also want to discuss the potential effects of predictors not included in the model (alpha = 0.05 is an arbitrary cut-off and a predictor with p = 0.06 still shows evidence of a (weak) association – different journals and statistical reviewers have different viewpoints on this - some journals/ reviewers don't want them included (perceived as 'fishing'/ data dredging)
- For backward elimination the coefficients of the predictor at the last step before it was eliminated can be reported
- Observational studies: often one Table presents univariate results and one Table shows the final multivariable model (if there are a lot and/or journal does not want non-significant univariate results in the manuscript the univariate Table(s) can be presented in an appendix/ online supplementary file

Workflow: Steps in Model Building

1. Experimental and Analytical Design
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Data/Inferential Analysis for Interpretable Models

Fit a single model LM1/2/3, Survival analysis

1. Check model assumptions
2. Check goodness-of-fit
3. Interpreting Model Parameters and reach a conclusion

Iterative Model building

- Pick predictors to fit and a suitable modelling method based on EDA
- Iteratively fit and assess models using predetermined strategy

5. Publication/Report

References used when building this Workshop

- Dohoo et al 2009 Veterinary Epidemiologic Research
- Rothman et al 2008 Modern Epidemiology, 3rd ed, Chapter 12
- Harrell, Frank E. Regression Modeling Strategies : with Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis . Second edition. Cham: Springer, 2015. Print. https://sydney.primo.exlibrisgroup.com/permalink/61USYD_INST/14vvljs/alma991008621859705106
- Steyerberg EW (2019) Clinical Prediction Models: A Practical Approach to Development, Validation, and Updating https://sydney.primo.exlibrisgroup.com/permalink/61USYD_INST/1367smt/cdi_askewsholts_vlebooks_9783030163990
- Quan H, Sundararajan V, Halfon P, Fong A, Burnand B, Luthi JC, et al. *Coding algorithms for defining comorbidities in ICD-9-CM and ICD-10 administrative data*. Medical Care 2005; 43(11):1130-1139. DOI: 10.1097/01.mlr.0000182534.19832.83

Model Building: Software resources

Linear model building example in R and other relevant resources for R:
https://r4ds.had.co.nz/model-building.html

Daggity: DAGitty - drawing and analyzing causal diagrams (DAGs)

Further assistance at The University of Sydney

SIH

- Statistical Resources website: containing our workshop slides and our favourite external resources (including links for learning R and SPSS).
- Hacky Hour: an informal monthly meetup for getting help with coding or using statistics software.
- 1on1 Consults can be requested on our website or here (click on the big red ‘contact us’ link).
- Online library. Useful links and the most recent version of all our workshops.

SIH Workshops

- Create your own custom programs tailored to your research needs by attending more of our Statistical Consulting workshops. Look for the statistics workshops on our training page or on our Training calendar.
- Sign up to our mailing list to be notified of upcoming training.

Other

- Linkedin Learning

A reminder: Acknowledging SIH

- All University of Sydney resources are available to researchers free of charge.

- The use of the SIH services including the Artemis HPC and associated support and training warrants acknowledgement in any publications, conference proceedings or posters describing work facilitated by these services.

- The continued acknowledgment of the use of SIH facilities ensures the sustainability of our services.

Suggested wording for use of workshops and workflows:

- *“The authors acknowledge the Statistical workshops and workflows provided by the Sydney Informatics Hub, a Core Research Facility of the University of Sydney.”*

We value your feedback

- We want to hear about you and whether this workshop has helped you in your research. What worked and what didn't work.

- We actively use the feedback to improve our workshops.

- Completing this survey really does help us and we would appreciate your help! It only takes a few minutes to complete (promise!)

- You will receive a link to the anonymous survey by email.

Appendix
DAG - a graphical aid to understand multivariable systems and relationships between variables

THE UNIVERSITY OF SYDNEY

What is a DAG?

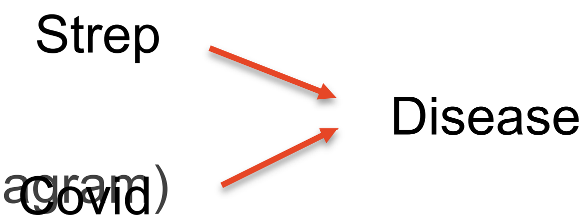
- DAG (Directed Acyclic Graph) = causal diagram = modified path model
- Start with a plausible (biological) causal structure (data generating process – DGP) and translate it into a graph with hypothesised and known relationships among variables
- Lines represent:
  - Arrow (directed edge) = (assumed) causal relationship
  - No arrow = no causal relationship

Example study aim: Identify factors of causal importance to pneumonia (lung disease)

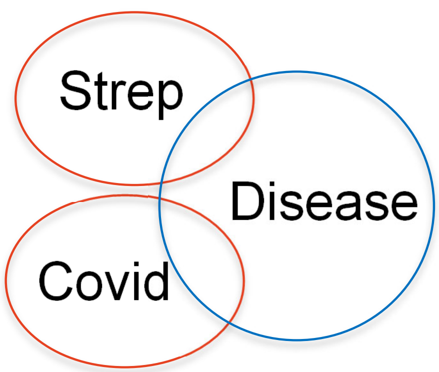
Objective: Investigate the association of Strep infection and the occurrence of lung disease. Researchers also measured sero-conversion for COVID-19.

Relationship: Exposure-independent predictor variable

Causal model (DAG)



Statistical model (Venn diagram)



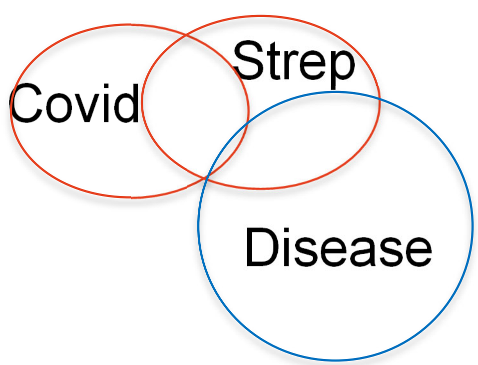
The two predictor circles do not overlap – they are statistically independent.
Both overlap with the outcome indicating their significant statistical associations with Disease.
A patient could have one infection, both infections or none.

Relationship: A simple antecedent predictor variable

Causal model (DAG)

Covid → Strep → Disease

Statistical model (Venn diagram)

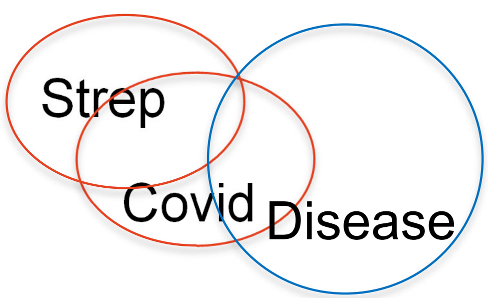


There is weak overlap of Covid with the outcome but statistical association favour direct causes over indirect causes, so strength of association and significance of Covid may be low. The association of Strep and Disease would not change when the model is adjusted for Covid. Covid occurs before Strep infection making patients more susceptible – Covid may be very important for disease control!

Relationship: An explanatory antecedent variable – complete confounding

Causal model (DAG)

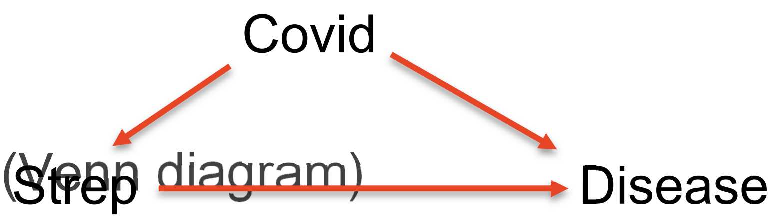
Statistical model (Venn diagram)



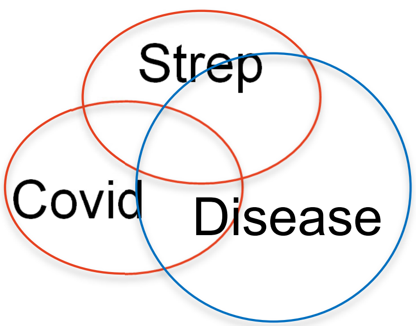
The Strep circle overlaps with the outcome, as they are statistically related until Covid is added to the model. Then the association becomes non-significant as all of the previous crude association is covered by the Covid-Disease association.

Relationship: an explanatory antecedent variable – partial confounding

Causal model (DAG)

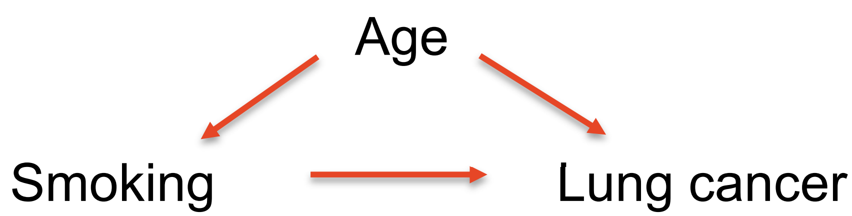


Statistical model (Venn diagram)



The Strep circle overlaps with the outcome. The association remains significant when Covid is added to the model but some of the previous association is now covered by the Covid-Disease association.
The Strep-Disease association is not as strong when Covid confounding is controlled but the model with both predictors explains more variation in Disease than just Strep.

More on confounding: a classic example



Age confounds the relationship of smoking and lung cancer – smoking may be more prevalent among older people and older people are more likely to get lung cancer irrespective of smoking.

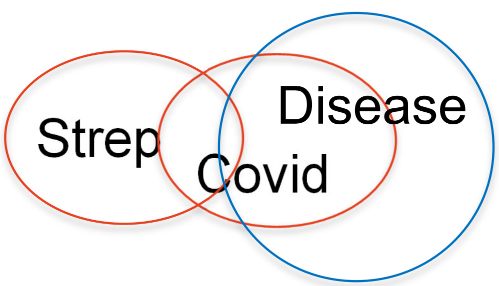
Examples of common confounders:
Age, Sex, Socio-economic status, Comorbidity, previous experience, etc.

Relationships: an intervening predictor variable

Causal model (DAG)

Strep → Covid → Disease

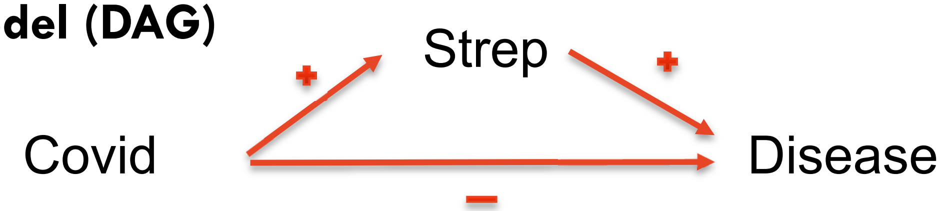
Statistical model (Venn diagram)



The Strep circle might or might not overlap with the outcome. However, any association of Strep with Disease disappears when Covid is added to the model. The effect of Strep on Disease is mediated by Covid. Adding Covid to the model would lead us to believe that Strep is not associated with Disease (and hence we may wrongly conclude that Strep is not a cause of Disease). Intervening variables should be identified and not be used/controlled when estimating the causal effect of an exposure → consider Structural equation modelling, mediation analysis instead if interested in the mediator.

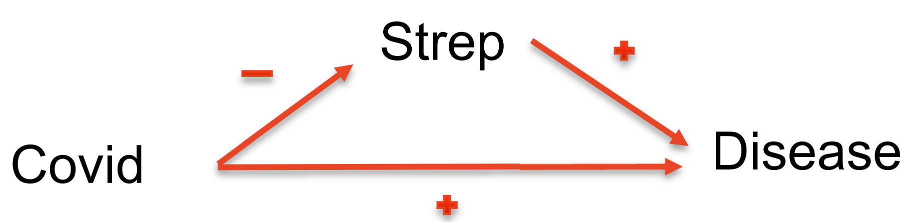
Relationship: a distorter predictor variable, can cause association reversal - one relationship represents prevention (-ve) rather than causal effect

Causal model (DAG)



Strep is a cause of Disease. Having Covid prevents getting Disease. Covid is positively correlated with Strep.

Causal model (DAG)



Strep and Covid are both causes of Disease. Having Covid prevents Strep infection.

In either case – need to control for the distorter variable Covid

Relationship: a suppressing predictor variable

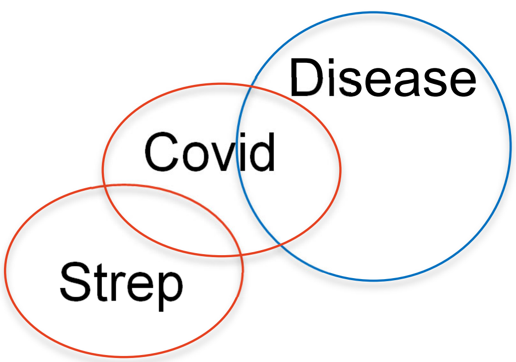
Causal model (DAG)

Hospital contact

Statistical model

$\boxed{\text{Strep} \\ \text{Covid}} \longrightarrow$ Disease

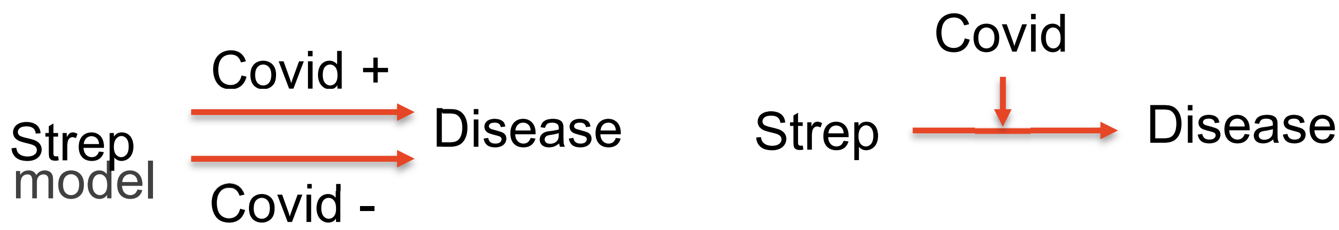
Strep and the suppressor Covid are both members of the same global variable ‘Hospital contact’ (a proxy variable for exposure to infectious agents)



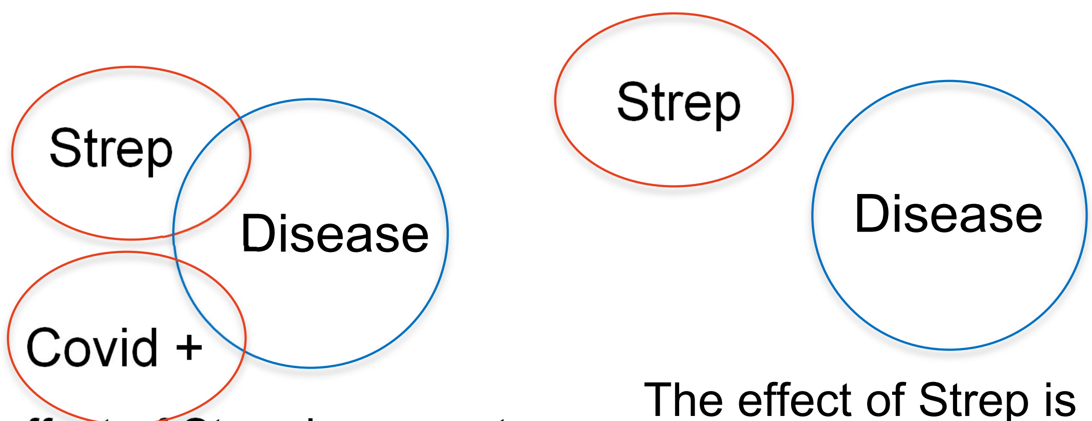
The variable ‘hospital contact’ has no or only weak unconditional association with the outcome. Once Covid is controlled in the analysis, the Strep circle overlaps with the outcome indicating an association of ‘hospital contact’ with Disease. By controlling the non-causal component of the global variable we reveal and strengthen the suppressed association of the remaining factor with the outcome. The global variable should be refined to exclude Covid. Suppression of the outcome can also occur, e.g. if the disease is not well defined (no association with lung disease but a strong association with a specific type of lung disease).

Relationship: a moderator variable – statistical interaction

Causal model (DAG)



Statistical model



The effect of Strep is present when Covid is present (Covid +)

The effect of Strep is absent when Covid is absent (Covid -)

The Strep circle overlaps with Disease only when Covid is present. No disease occurs unless both predictors are present. Interaction has large implications for disease control. Moderators may or may not be confounders.

Summary of effects of extraneous variables

Covid is a(n) … variable	Likely effect on Strep coefficient when adding Covid	Comments
Exposure independent	No change	Covid explains some of the Disease incidence, so the residual variance is smaller and significance of Strep increases.
Simple antecedent	No change	No effect on the analysis by Covid, but Covid might be important to know about from a prevention perspective, e.g. if easier to address than Strep.
Explanatory antecedent (complete confounding)	Becomes 0	Control of Covid will remove any Strep association with Disease. R squared should increase as residual variance decreases.
Explanatory antecedent (incomplete confounding)		Controlling Covid will impact on significance of Strep depending on the strength of Covid effect on Strep and disease. R squared should increase.
Intervening		Because Covid is more closely related with Disease it probably has a stronger association and explains more variability. Strep coefficient is reduced in size and significance. If all effect passes through intervenor it will remove Strep effect on Disease.
Distorter		Essentially same impact as explanatory antecedent except the Strep effect is increased or in opposite direction to the crude association.
Suppresor		As the global variable containing Strep is refined, it will now have a stronger relationship with Disease and probably explain more variation.

RCT DAG – what is an ‘instrumental’ variable?

Example: A randomised Controlled Trial (RCT) experiment

Vaccination → Exposure to Covid virus → Covid infection

Treatment:
Control group – not vaccinated
Treatment group – vaccinated

Vaccination is an ‘**Instrumental variable**’, it has direct causal effect on exposure, is unrelated to the outcome and shares no common cause with the outcome